

Oriolowo Mustapha

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Education

Federal University of Agriculture, Abeokuta (FUNAAB)

Bachelor of Science, Computer Science

Abeokuta, Nigeria

Expected Aug 2028

Technical Skills

Languages: TypeScript, JavaScript, C#, SQL, Python, HTML5, CSS3

Frontend: React, Tailwind CSS, Vite, Zustand, TanStack Query, React Hook Form, Zod

Backend: Node.js, Express.js, ASP.NET Core Web API, Entity Framework Core, MediatR, FluentValidation, Mongoose

Architecture: Clean Architecture, CQRS, Transactional Outbox, REST API Design

Data & Infrastructure: PostgreSQL, MongoDB, Redis, BullMQ, Hangfire, Docker, Git, Swagger

Cloud & Testing: Azure (App Service, Virtual Machines, Application Insights), Railway, Vercel, Docker, Serilog, xUnit

Integrations & Security: Paystack, Nomba, OpenAI, Gemini, Groq, WhatsApp (OpenWA), JWT, RBAC

Work Experience

Sysbeams / Code Learners Hub

Abeokuta, Nigeria

Software Engineer (Intern & Fellow)

- Engineered secure REST APIs with C#, ASP.NET Core, and PostgreSQL across multiple enterprise platforms.
- Built payment processing modules for fee disbursements, integrating bank payment gateways and preventing duplicate submissions with idempotency keys.
- Designed relational schemas, stored procedures, and LINQ queries in PostgreSQL to maintain transactional consistency.
- Implemented multi-tier role-based access control (RBAC), cryptographic verification, and authentication layers.

Projects

InstaSafe | C#, .NET 10, ASP.NET Core, PostgreSQL, MediatR, Paystack, Azure

Solo Backend Engineer

- Designed and built the full backend of a production escrow platform for Nigerian social commerce that holds buyer payments until delivery is confirmed, then pays out vendors.
- Used Clean Architecture with CQRS (MediatR) and EF Core/PostgreSQL; wrote 222 passing tests across unit, domain, and integration suites.
- Integrated Paystack (collections, transfers, refunds, bank resolution), Groq, OpenWA, and Brevo, with a transactional outbox so notifications still arrive during outages.
- Added an AI WhatsApp chatbot for order creation where the LLM only proposes changes and all writes go through the same validators.
- Deployed the API to Azure App Service and self-hosted the OpenWA WhatsApp gateway on an Azure VM.

AgroGuardian AI | TypeScript, Node.js, Express, React, MongoDB, Redis, BullMQ, Gemini/OpenAI

- Built a full-stack climate-smart agriculture platform for AI crop and livestock diagnostics, climate risk modeling, and carbon-credit verification.
- Backend: BullMQ workers on Redis for diagnostics, weather syncs, and reminders; Gemini and GPT-4o with automatic fallback, Zod schema enforcement, and JSON repair.
- Built a carbon MRV engine that checks additionality rules and EXIF GPS/timestamp photo evidence, and a 5-factor climate risk model weighted by soil and irrigation.
- Frontend: React 19, Vite, TanStack Query, and Zustand with an async diagnostic polling flow, a 7-day risk view, and livestock management; Axios interceptors handle JWT auth and Google OAuth.

AjoCore | C#, .NET 10, ASP.NET Core, PostgreSQL, MediatR, Hangfire, Nomba API, Railway

DevCareer x Nomba Hackathon

- Designed and built a fintech backend that digitizes Ajo/Esusu cooperative savings (rotating, accumulated, and personal cycles) using Clean Architecture and CQRS.
- Provisioned Nomba virtual accounts per member and recorded contributions through HMAC-verified, idempotent webhooks; auto-reversed mismatched amounts.
- Built Hangfire jobs for daily payouts by bank transfer and contribution reminders; secured three roles with JWT refresh rotation.
- Containerized with Docker and deployed to Railway with managed PostgreSQL.

Coding Sandbox Platform | React, TypeScript, Zustand, C#, .NET, CQRS, Judge0

- Building an assessment platform that runs candidates' C# code in a Judge0-backed sandbox, with a React/TypeScript client and a CQRS .NET backend.

Skill Matrix 2.0 | C#, ASP.NET Core 8, PostgreSQL, MediatR, Groq AI, Judge0, Hangfire, React

- Built a multi-tenant, AI-powered skill assessment and gamification platform on Clean Architecture with CQRS; four RBAC roles with JWT and Google OAuth.
- Integrated the Groq LLM for timed assessments, skill-gap scoring, and 4-week improvement plans, plus a Judge0 code-judging pipeline for 6 languages.
- Implemented XP, streaks, badges, leaderboards, and endorsements; Hangfire sends weekly email reminders and an LMS webhook is secured by shared secret.
- Built the React manager dashboard mapping team capabilities with search, filtering, and sorting; redesigning it in a neumorphic style with gamification.

Awards

- 2nd Runner Up, GSpark Agritech Hackathon** — Won the NGN 200,000 grand prize and an NGN 100,000 scholarship at CodeLearnersHub for building AgroGuardian AI.